Trust Me, I'm a Certificate Authority

PyCon PT 2026 Tutorial

Tiago Montes

DISCLAIMER

THIS IS TEACHING MATERIAL, NOT SECURITY SOFTWARE.

IT CUTS CORNERS ON PURPOSE.

SEVERAL EXERCISES ARE DELIBERATELY BROKEN.

REAL SECURITY IS A GREAT DEAL MORE THAN THIS.

USE NONE OF IT FOR ANYTHING THAT MATTERS.

NEVER TRUST A CA YOU BUILT HERE.

Trust Me, I'm a Certificate Authority

- **RSA** Keys, encryption, and signatures.
- **Certificates** Binding keys to identities (yay X.509)
- **Cert Authorities** Becoming a Certificate Authority
- **TLS with HTTPS** Real HTTPS with a client that trusts.
- **MITM** Interception: where trust actually lives
- **Break Stuff** Certificates that fail, checks that catch them

RSA (Rivest–Shamir–Adleman)

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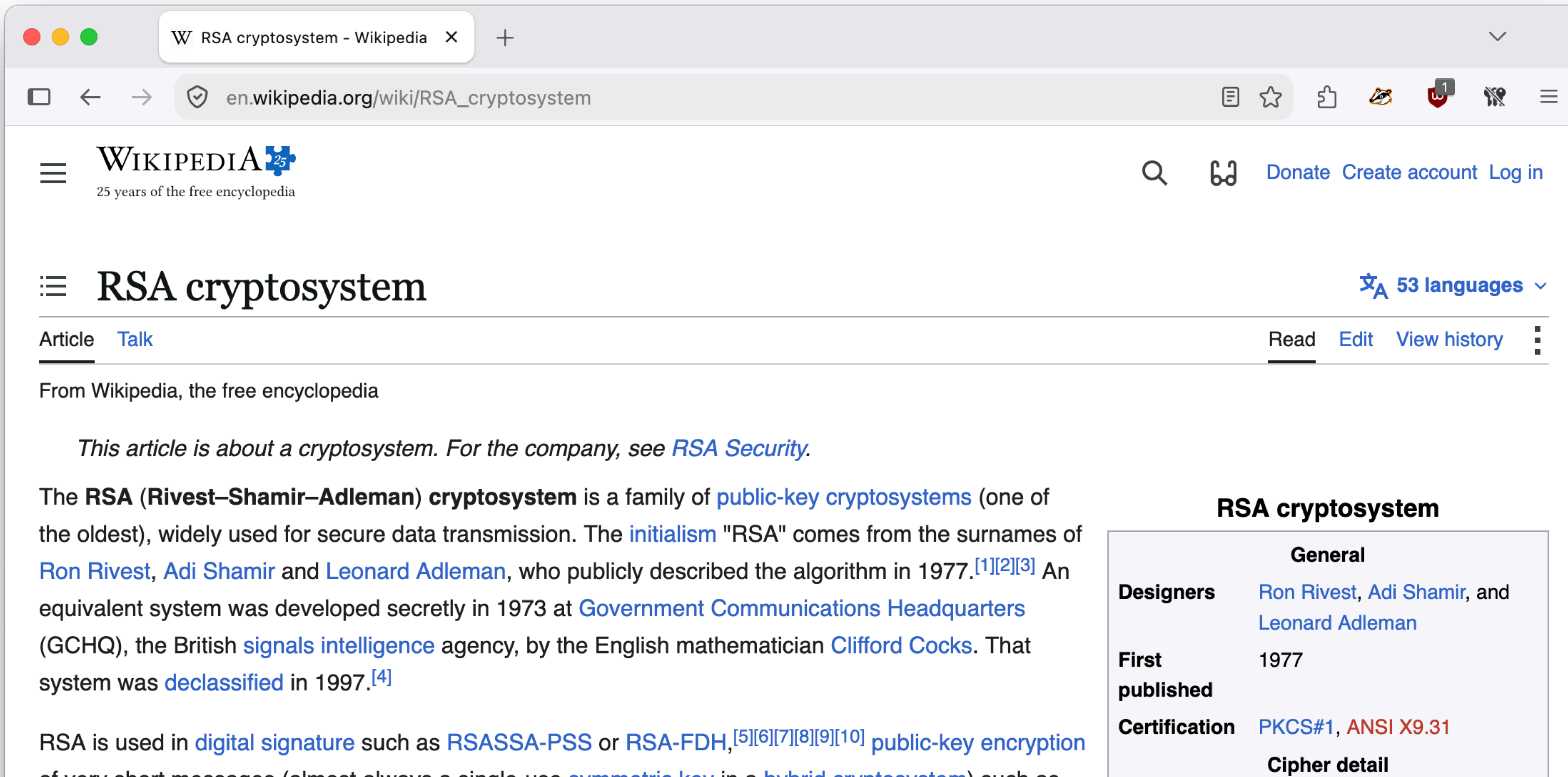
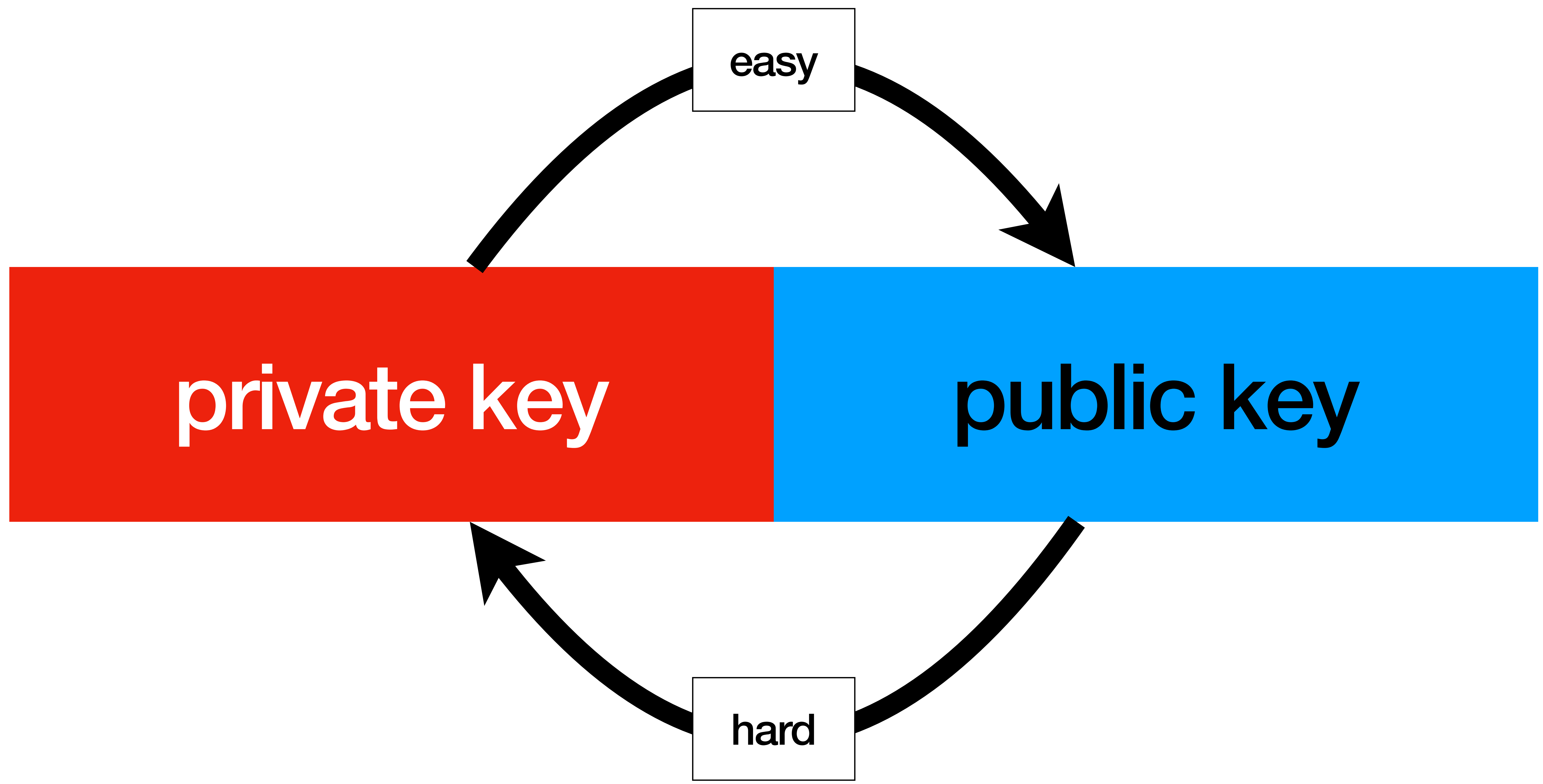


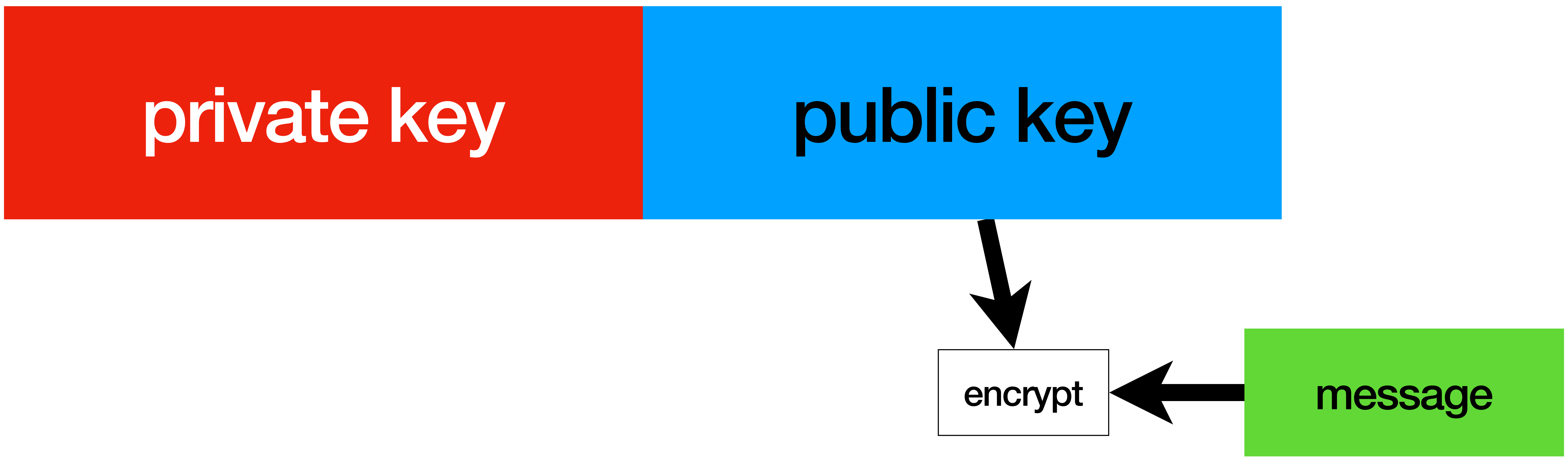
RSA key pair

private key | public key

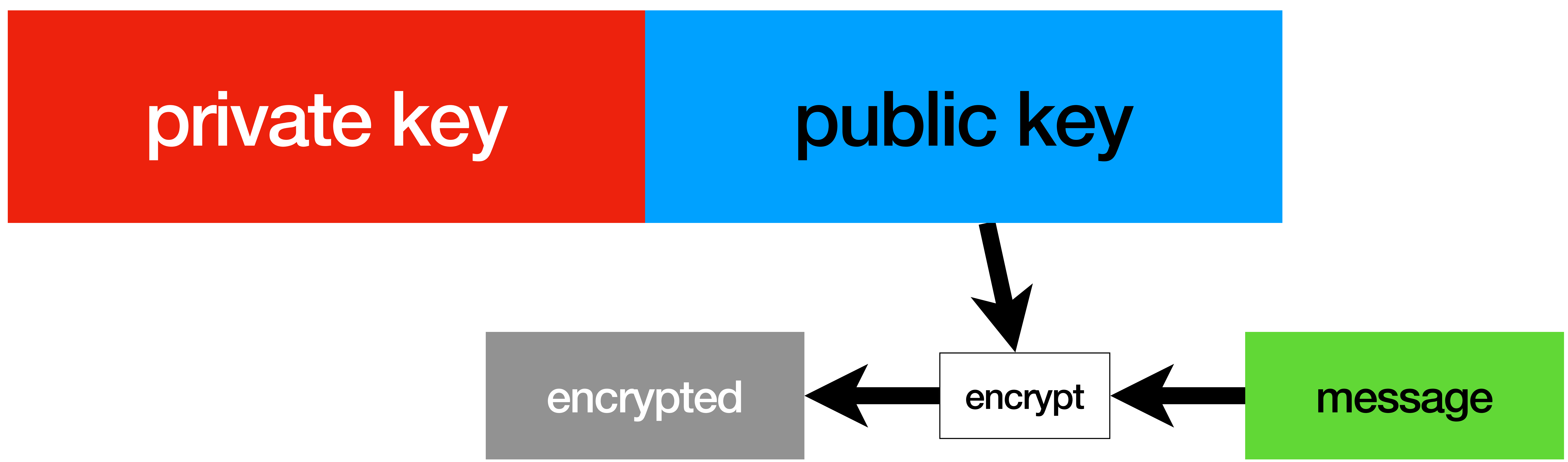
RSA key pair



RSA operations



RSA operations



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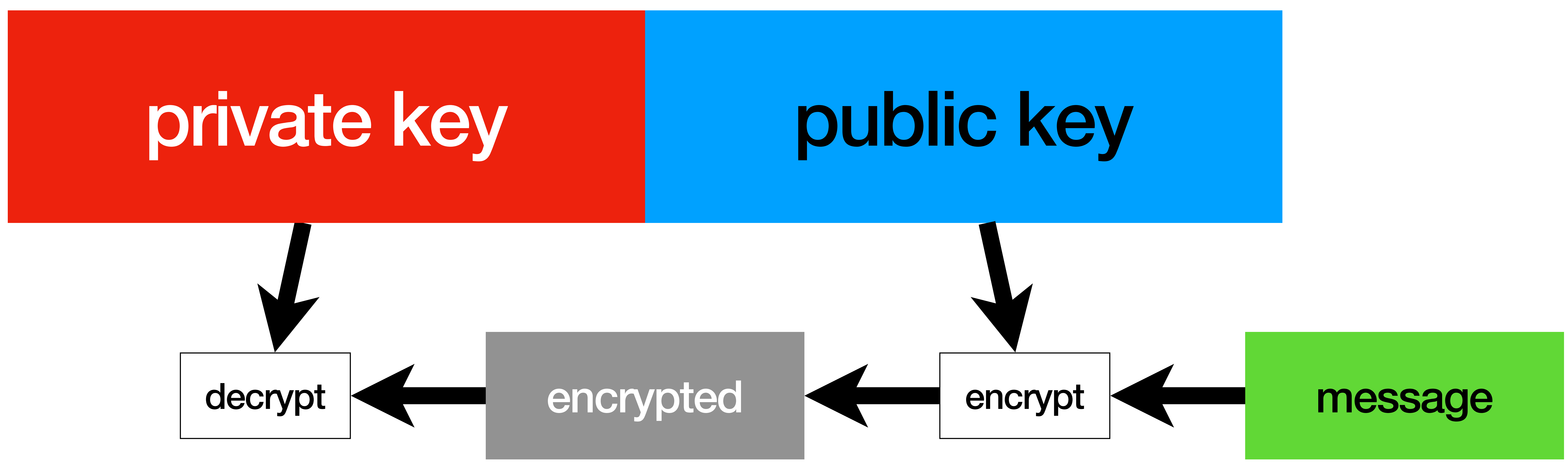

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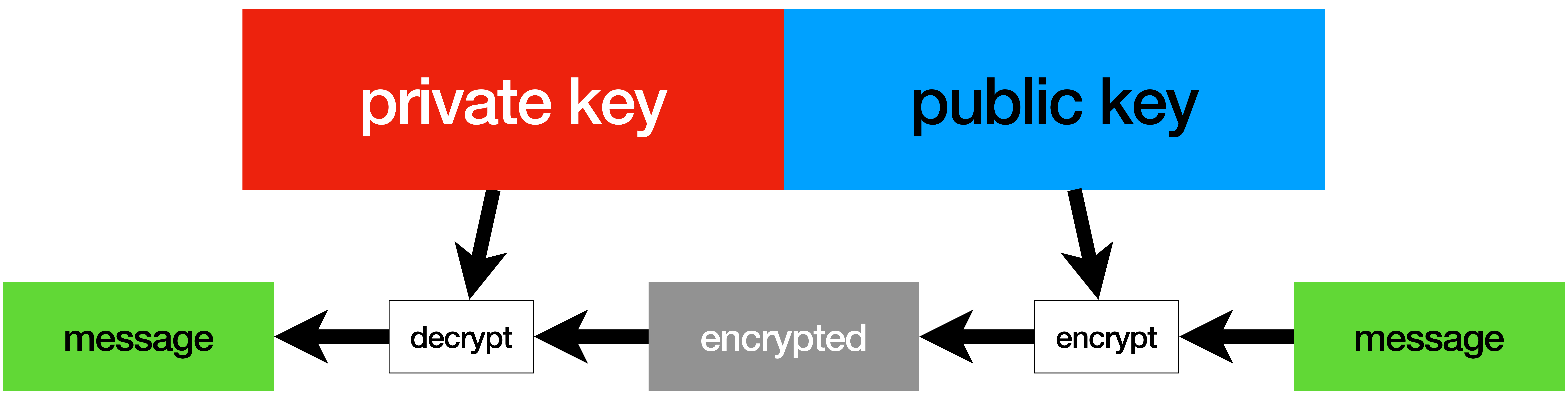
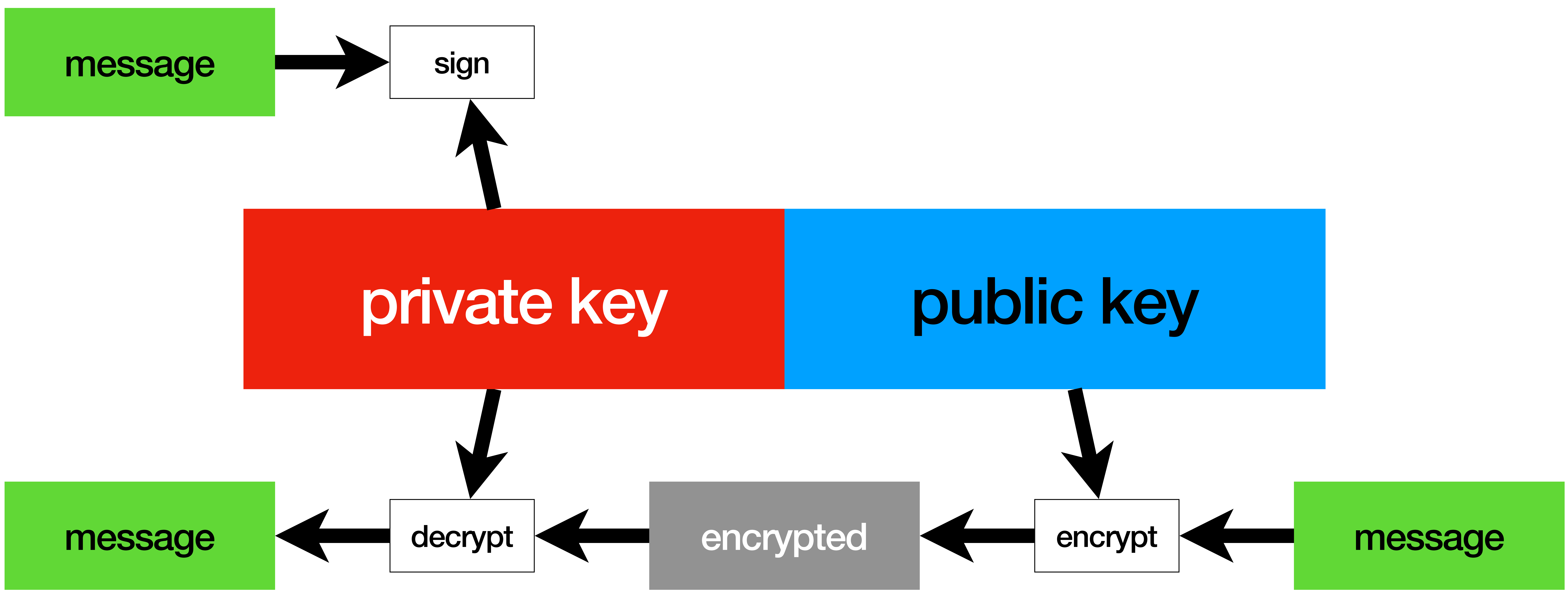


RSA operations



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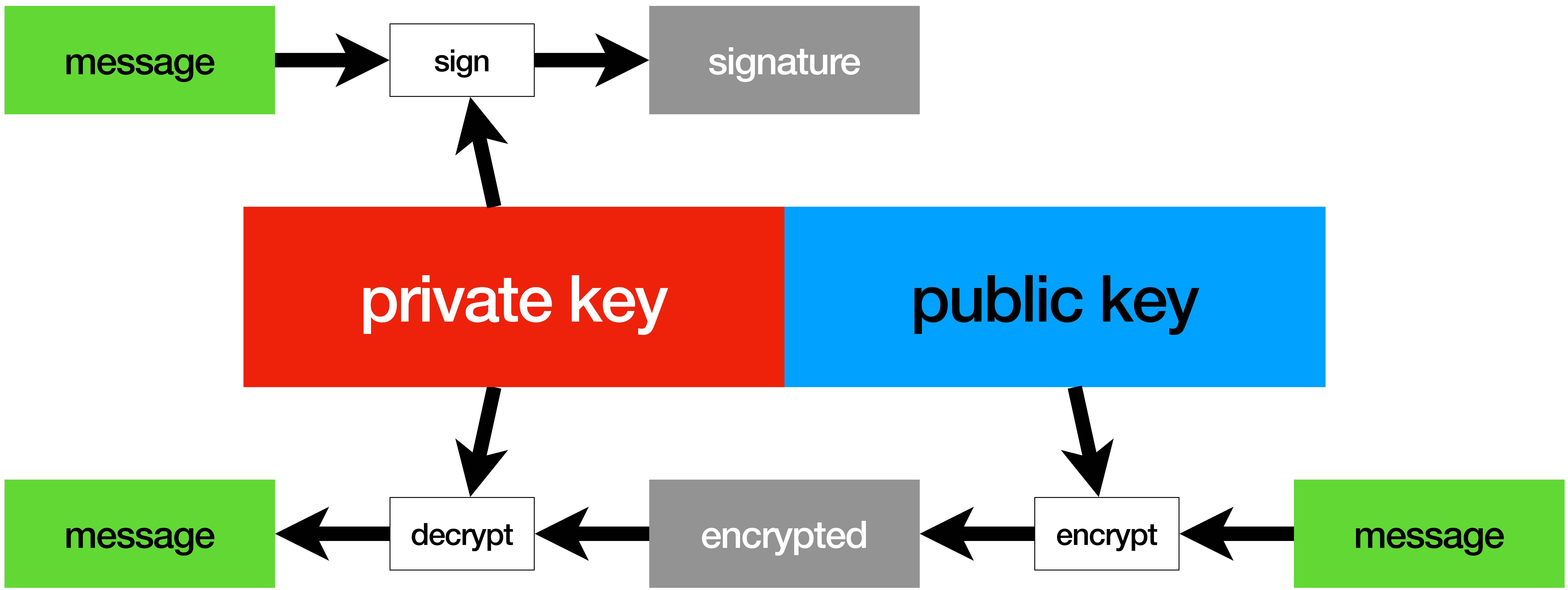

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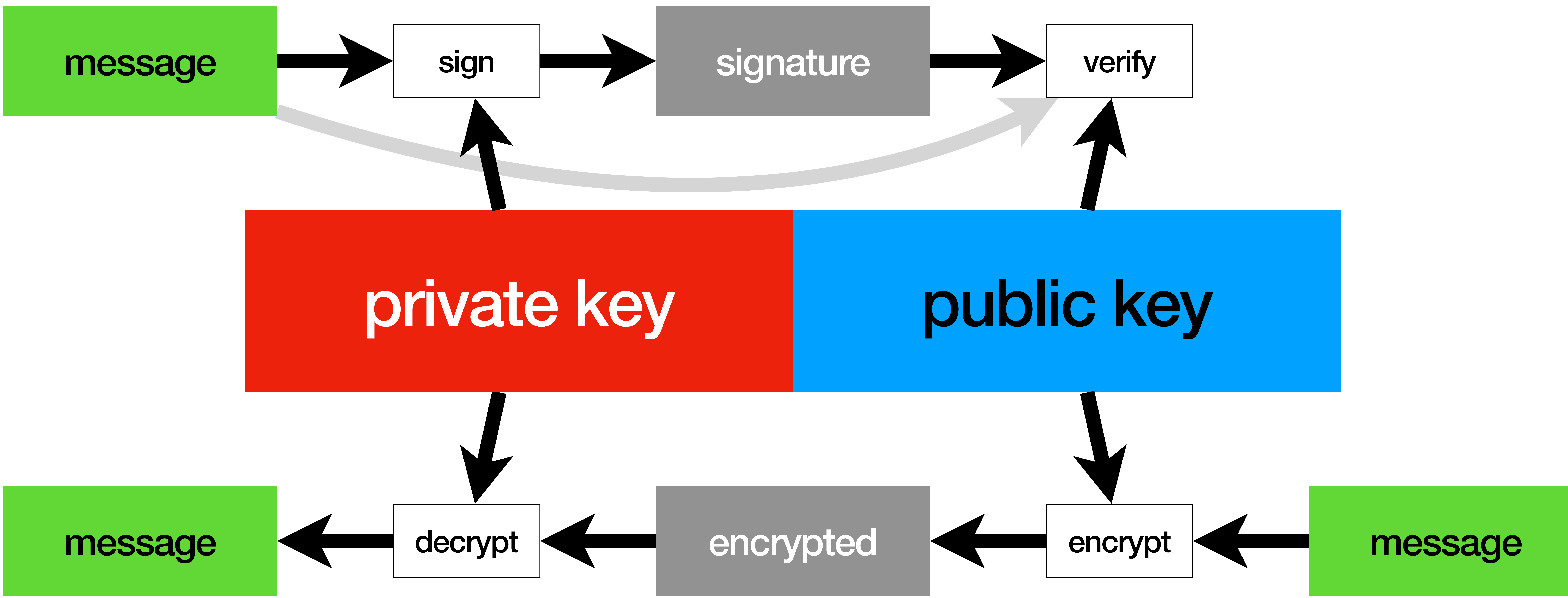

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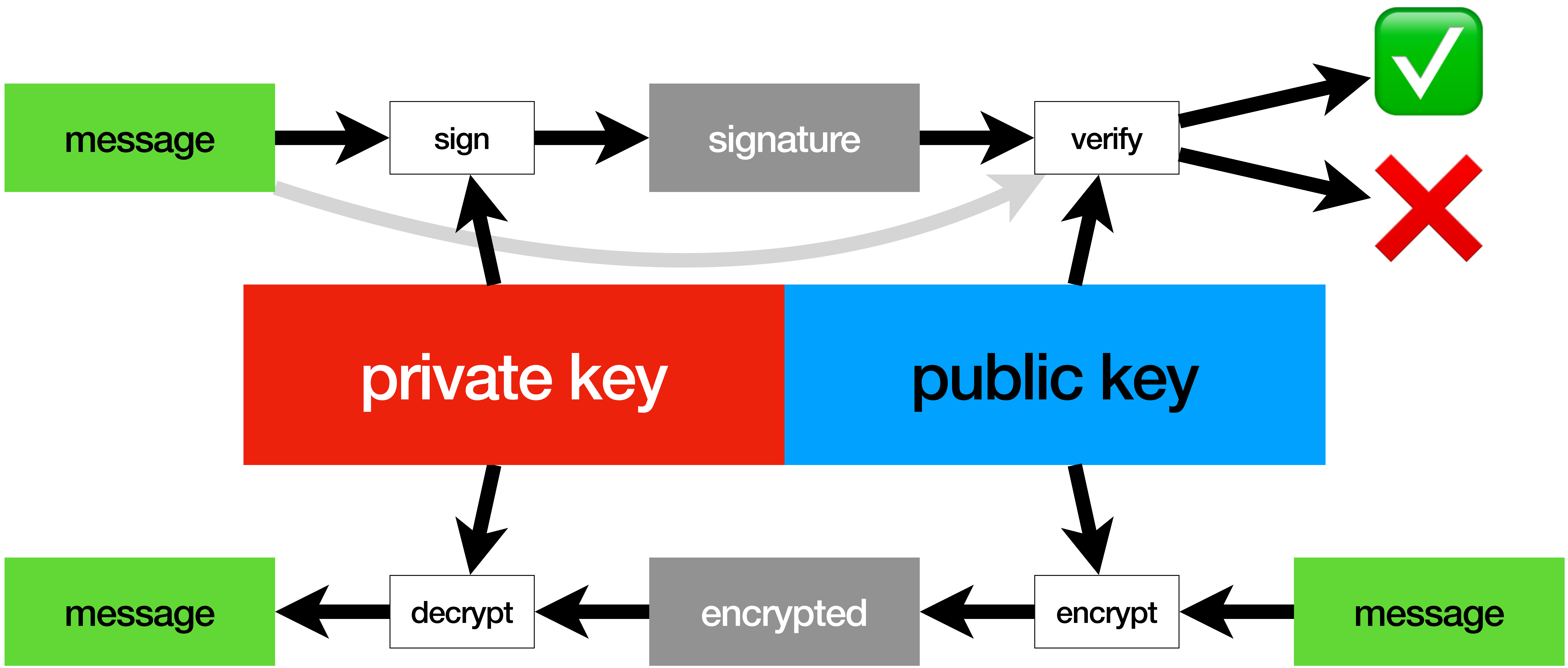

RSA private communication

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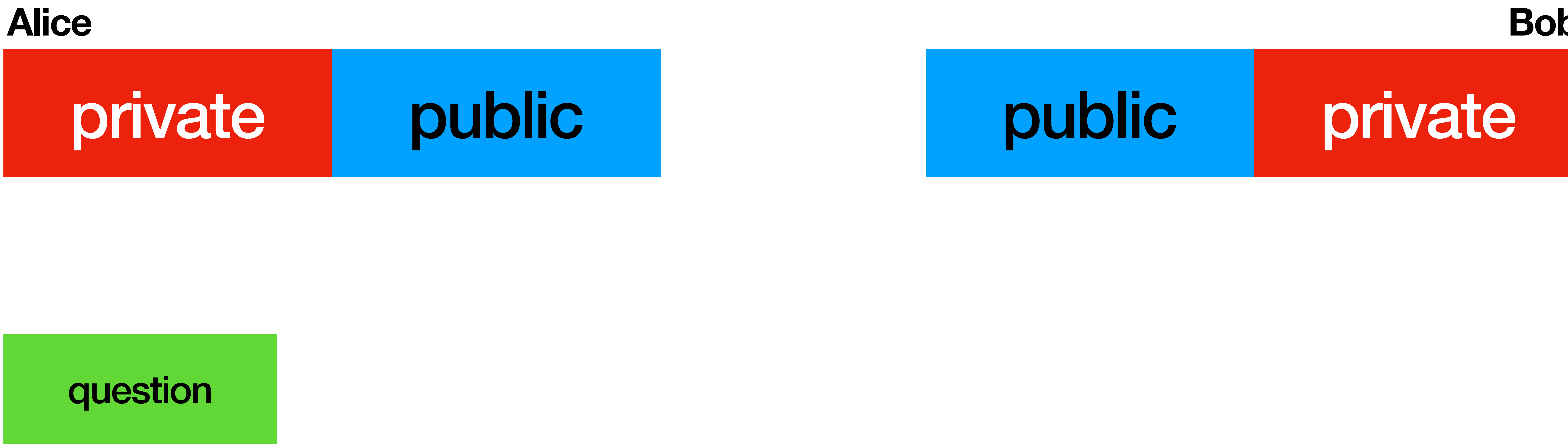
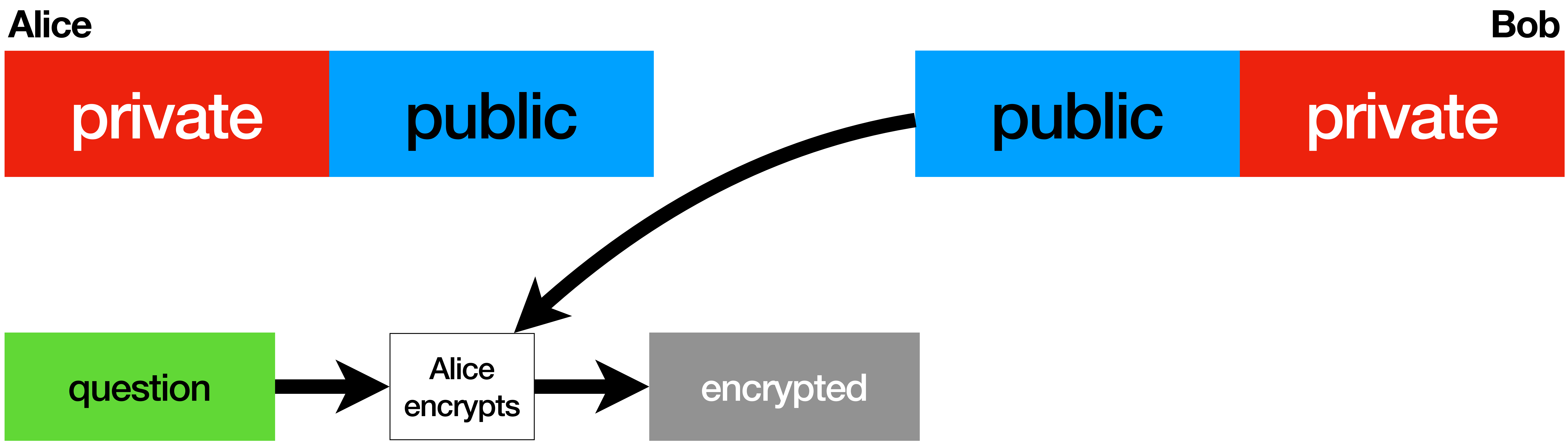


Alice sending a private question to **Bob**

RSA private communication



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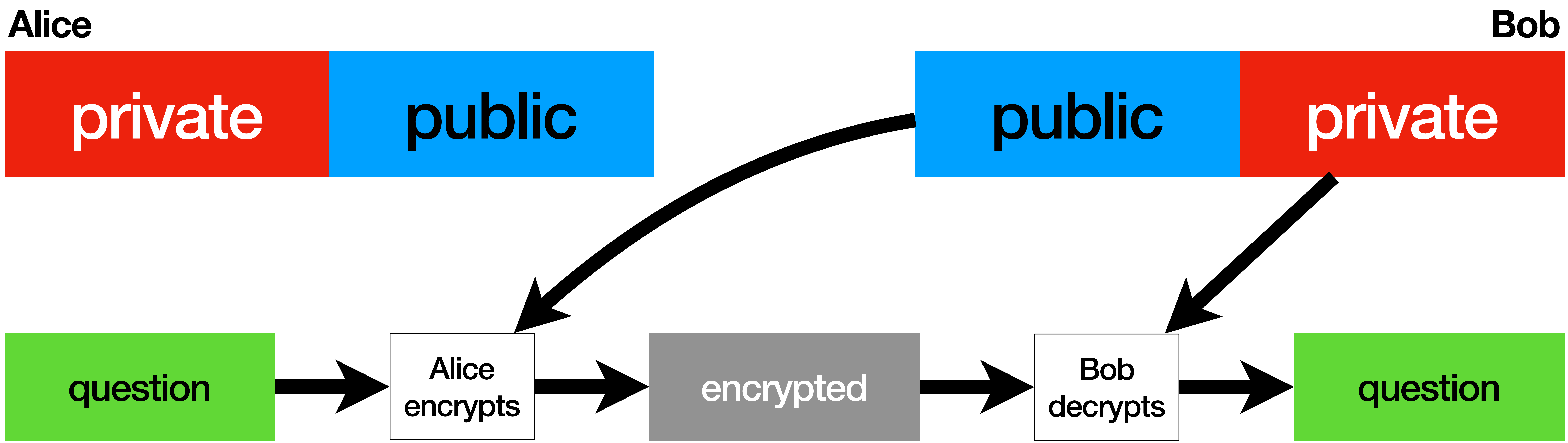

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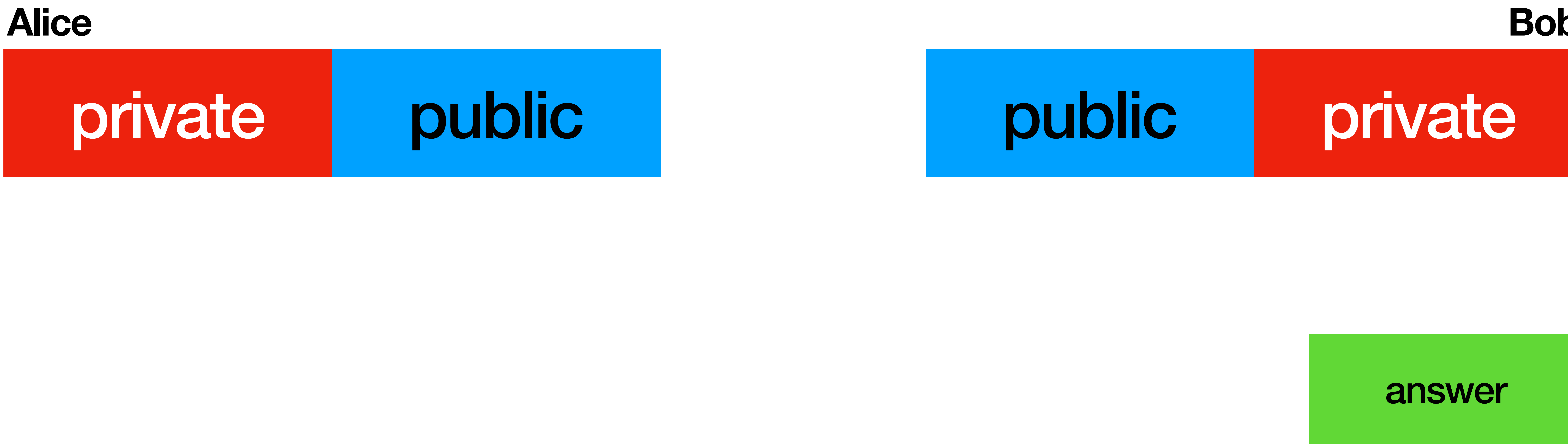
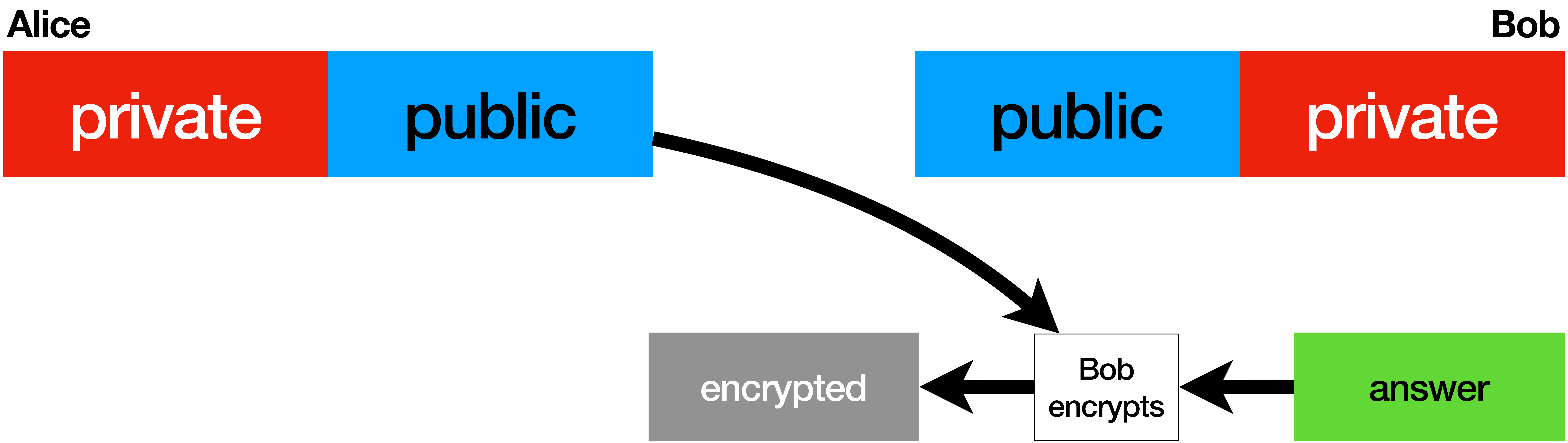


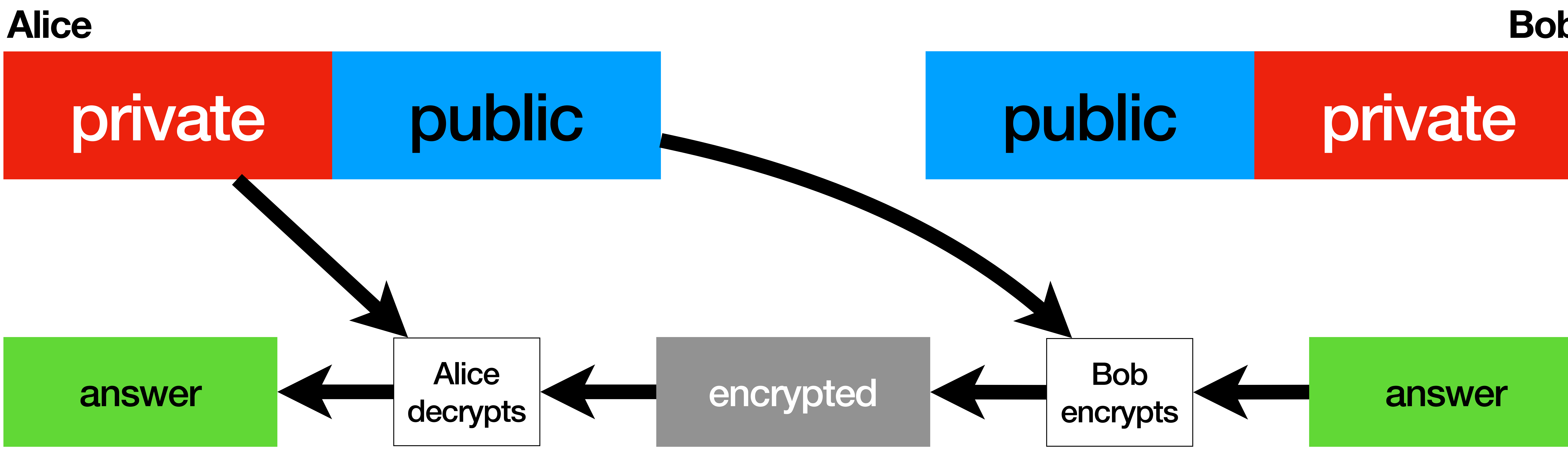
Bob sending a private answer to **Alice**

RSA private communication



Bob sending a private answer to **Alice**

RSA private communication



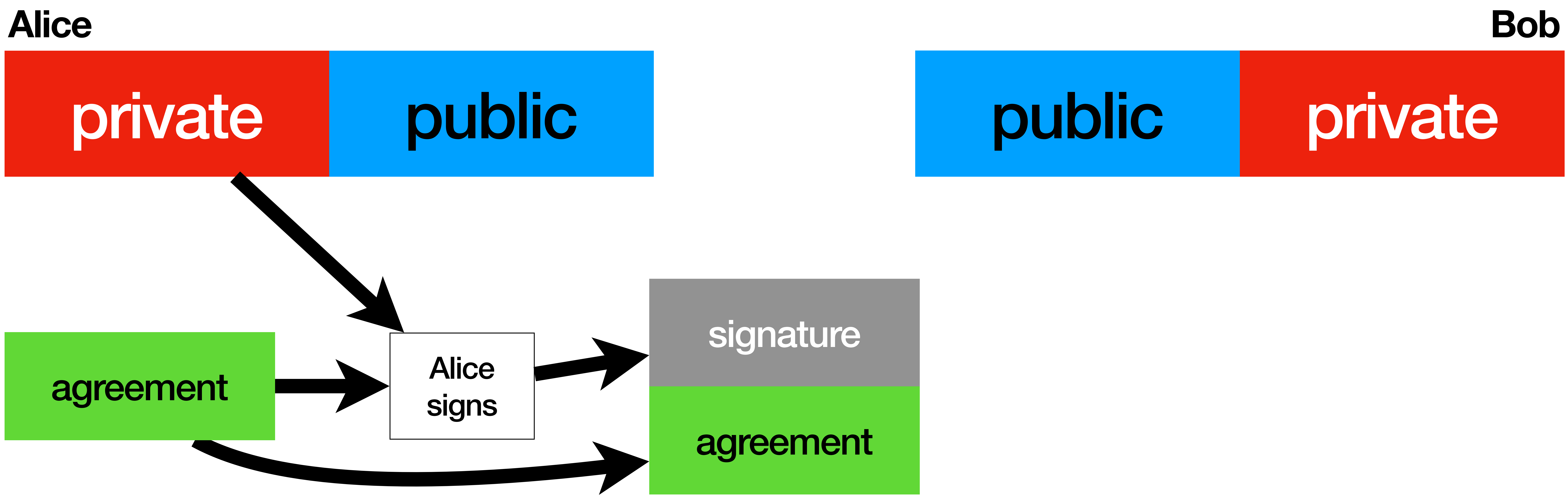
Bob sending a private answer to **Alice**

RSA signing / authentication

RSA signing / authentication

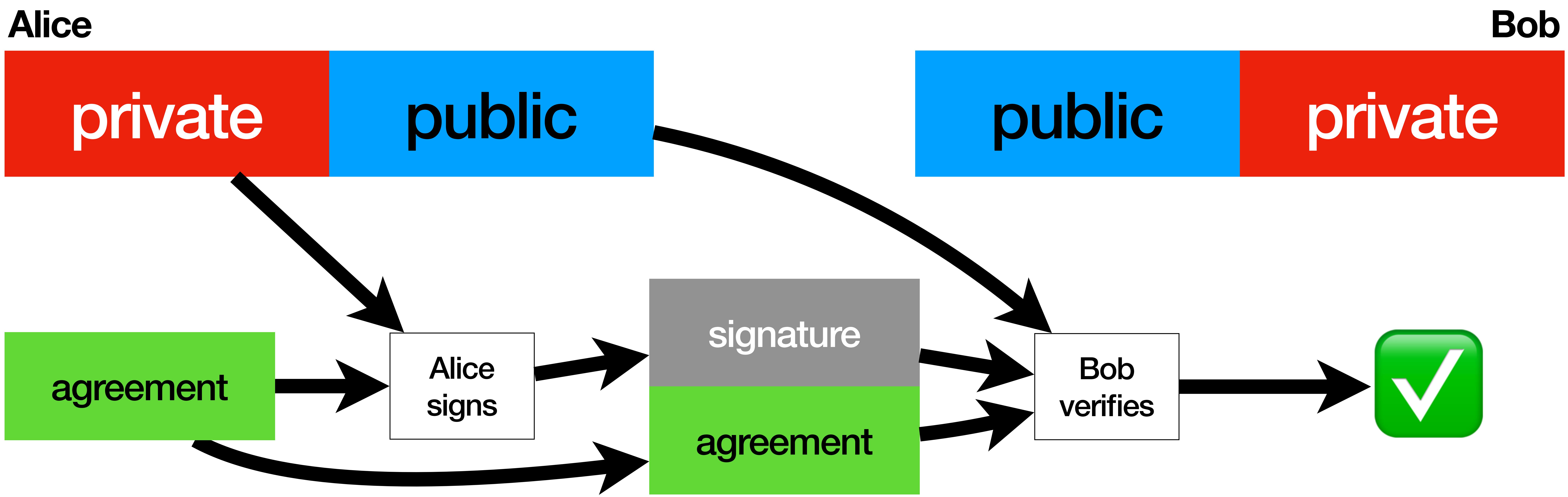
Alice sending an authenticated agreement to **Bob**

RSA signing / authentication



Alice sending an authenticated agreement to **Bob**

RSA signing / authentication



Alice sending an authenticated agreement to **Bob**

RSA private and authenticated

RSA private and authenticated

Alice

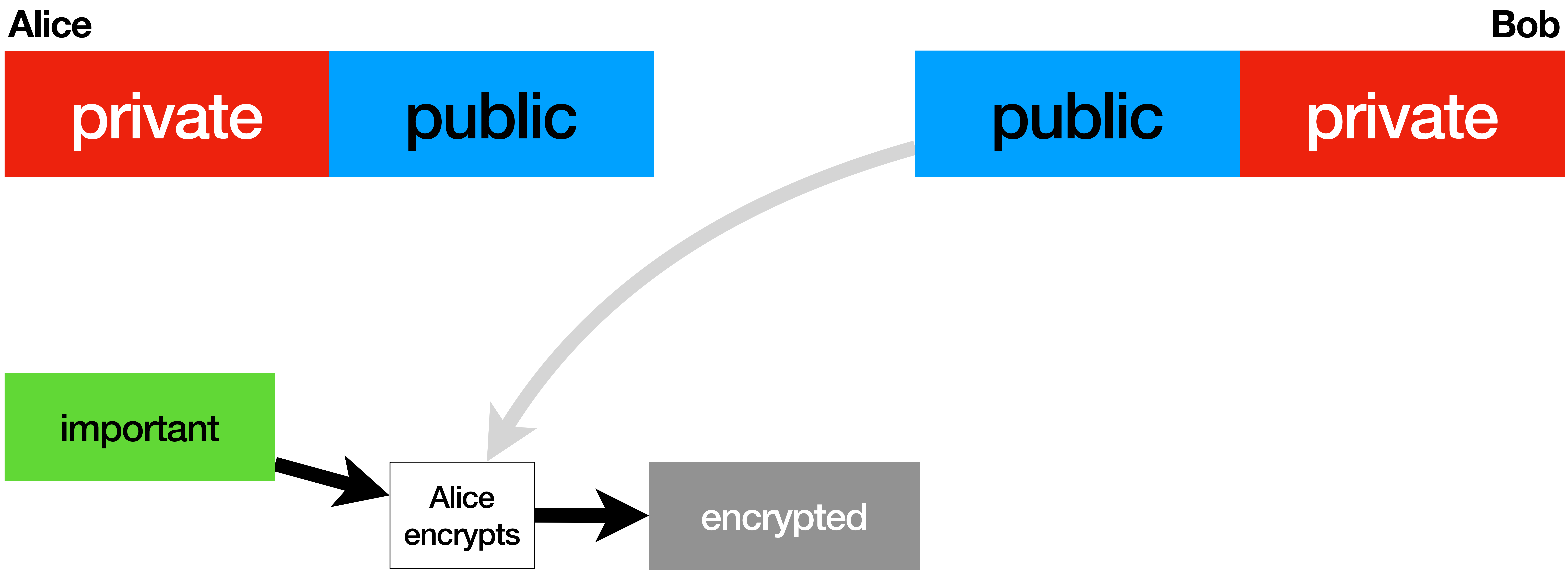
private public

Bob

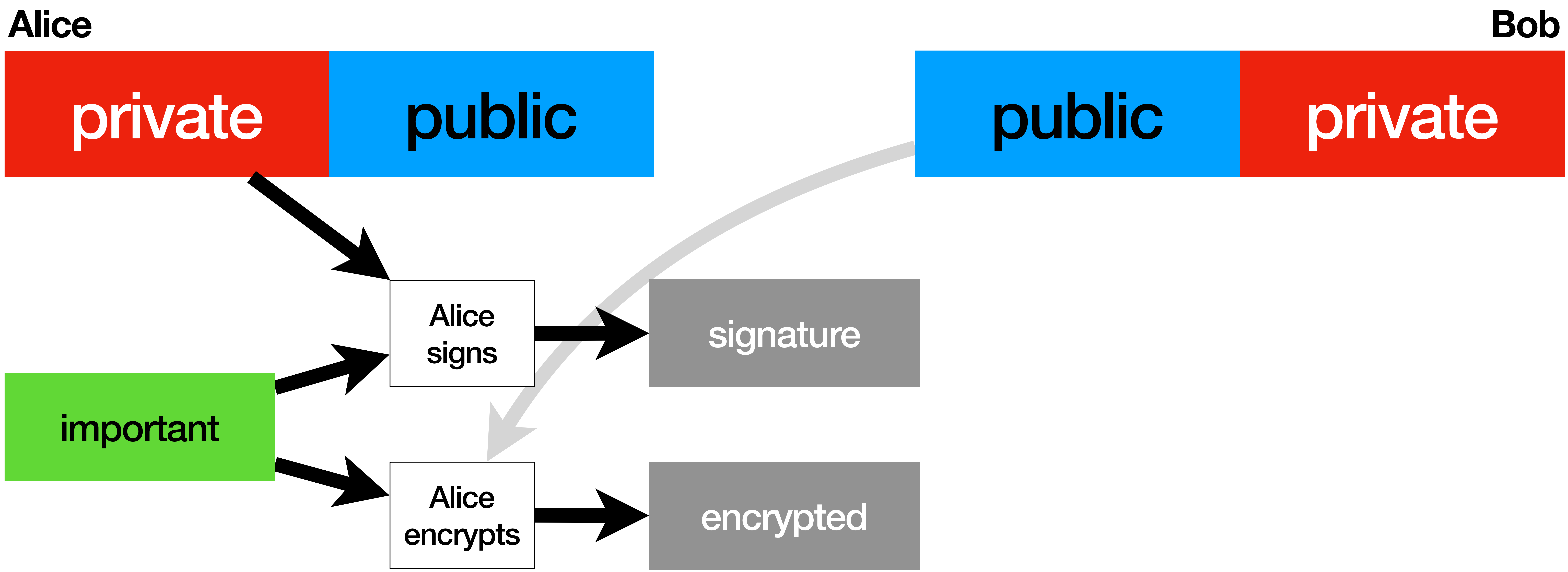
public private

important

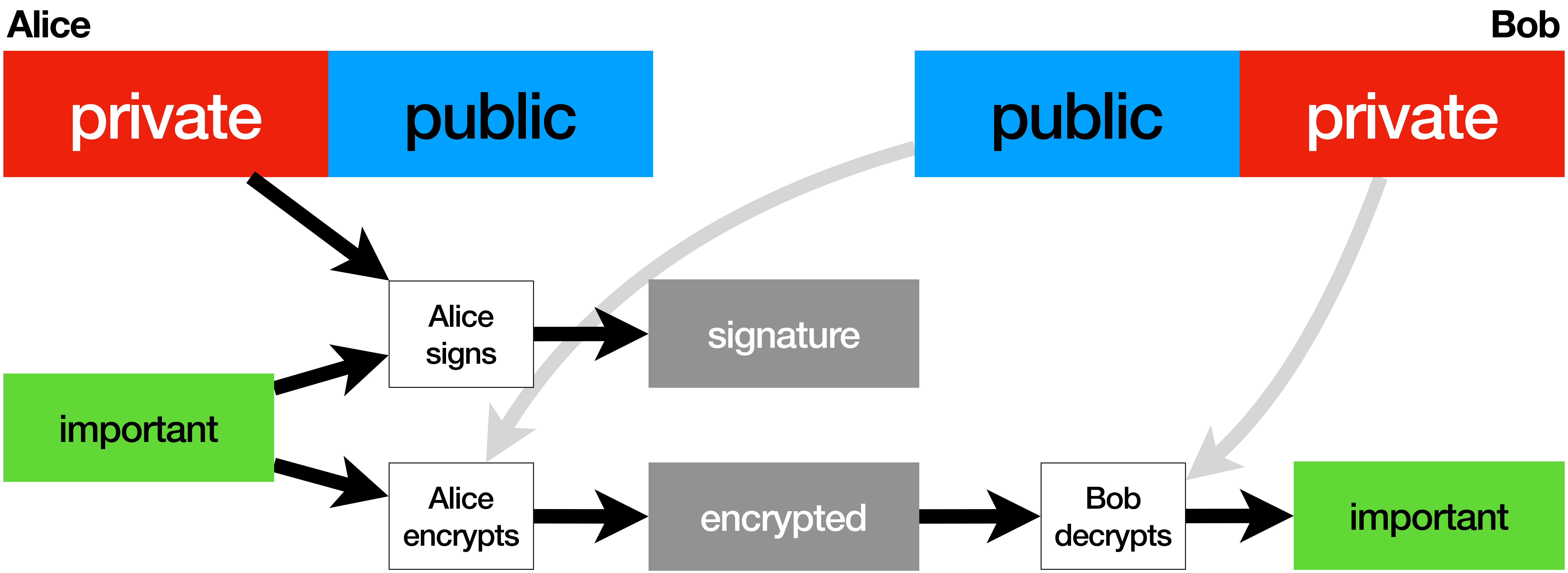
RSA private and authenticated



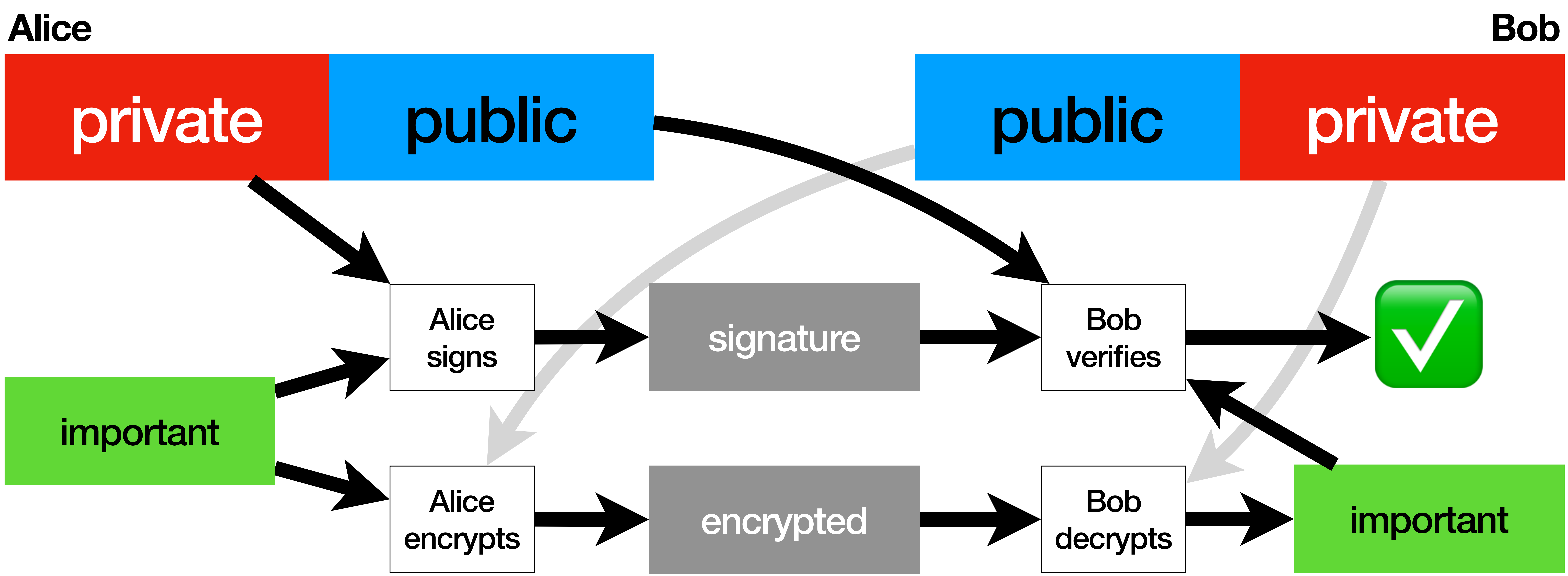
RSA private and authenticated



RSA private and authenticated



RSA private and authenticated



RSA

RSA

no way this is enough

X.509 Certificates

X.509 Certificates

issuer name
subject name
valid from
valid to
public key
signature

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issuer name
subject name
valid from
valid to
public key
signature

Issuer certifies,
by **signing** the document:

- This **public key**
- Belongs to this **Subject**
- During this **period**

X.509 Certificates

issuer name
subject name
valid from
valid to
public key *the subject's*
signature *the issuer's*

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X.509 Certificates

issuer name
subject name
valid from
valid to
public key *the subject's*
signature *the issuer's*

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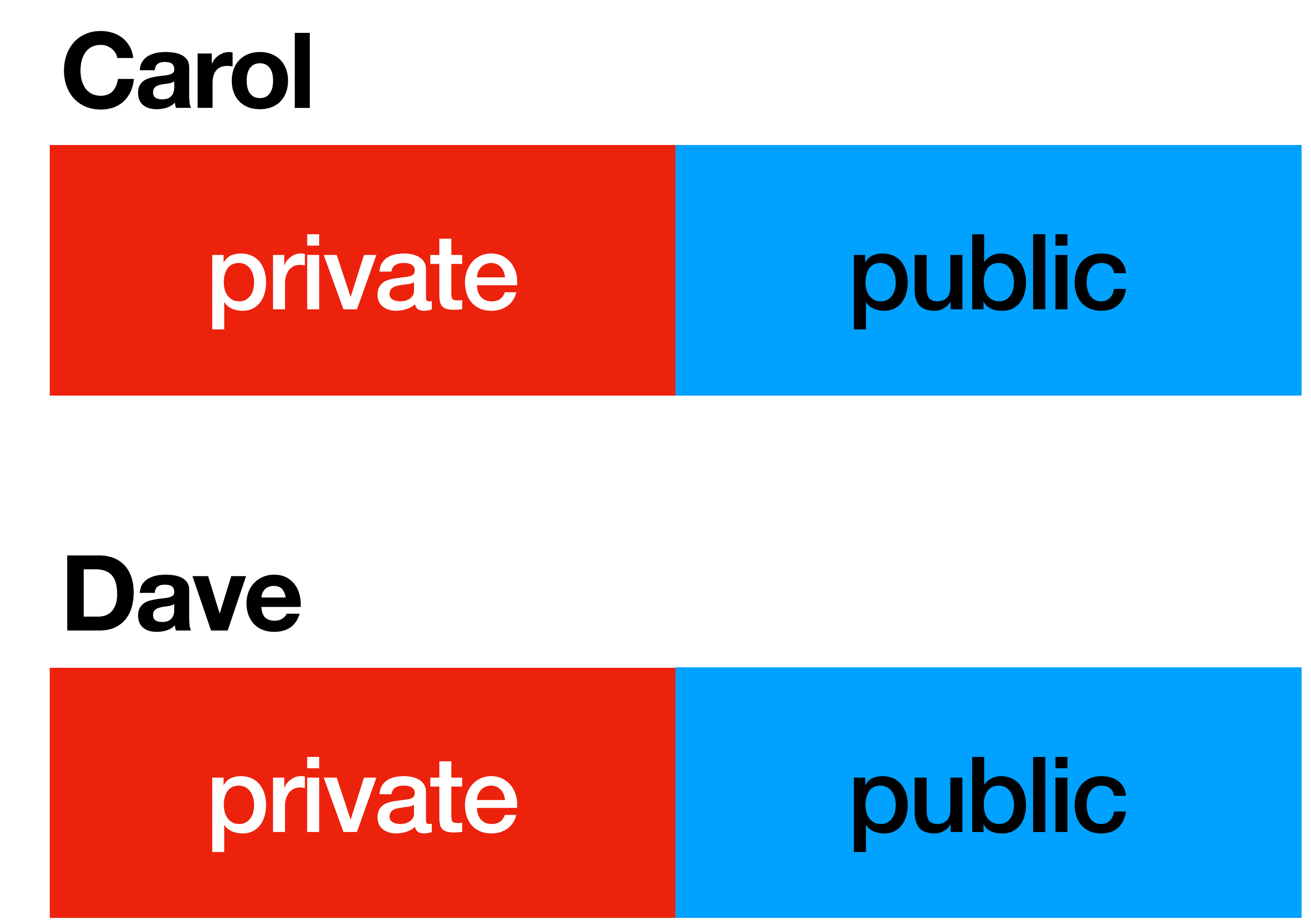
Certificates provide **Identity Assurance**

X.509 Certificates



Carol certifying this public key is **Dave's**

X.509 Certificates

issuer name	Carol
subject name	Dave
valid from	...
valid to	...
public key	
signature	

Carol

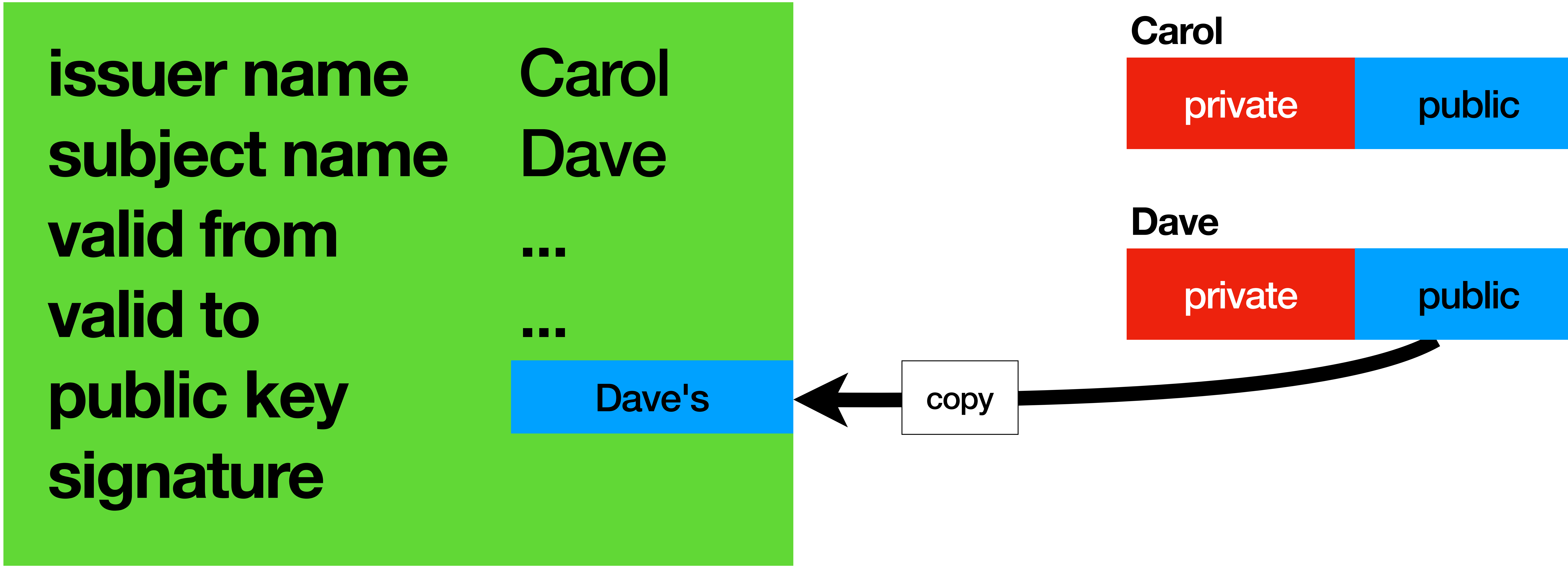
private	public

Dave

private	public

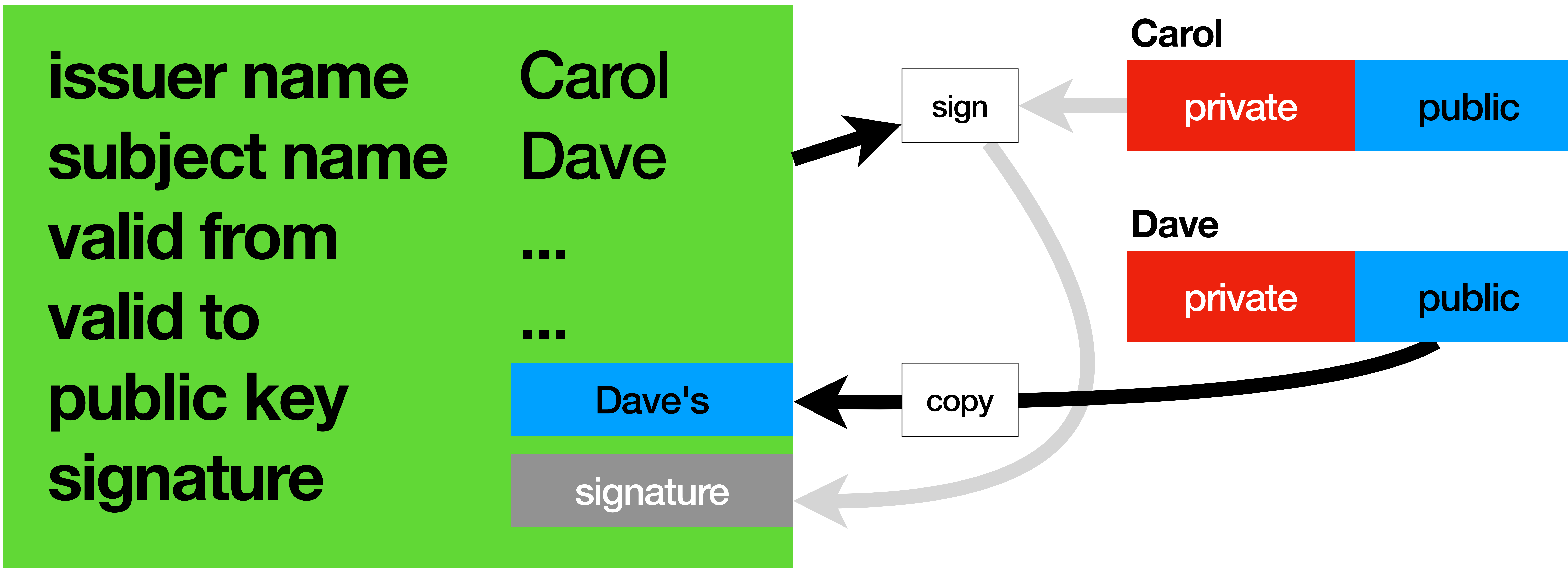
Carol certifying this public key is **Dave's**

X.509 Certificates



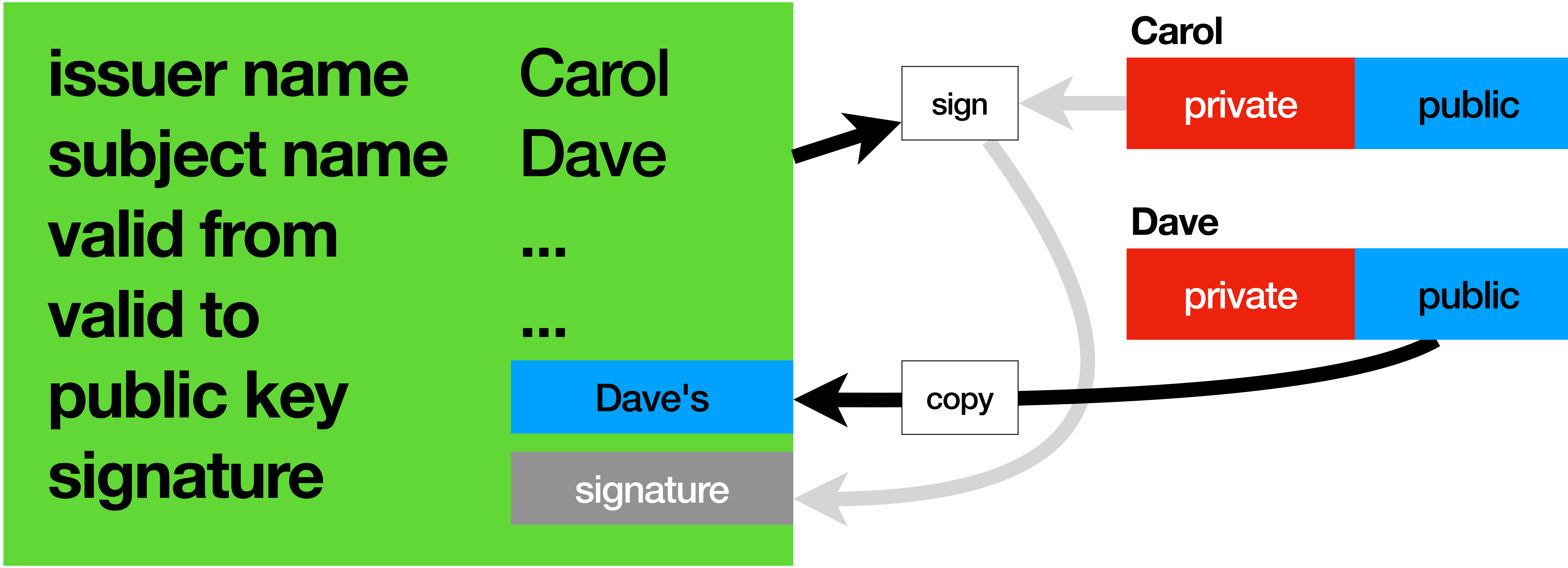
Carol certifying this public key is **Dave's**

X.509 Certificates



Carol certifying this public key is **Dave's**

X.509 Certificates



Certificates hold no secrets and are **public.**

X.509 Self-Signed Certificates

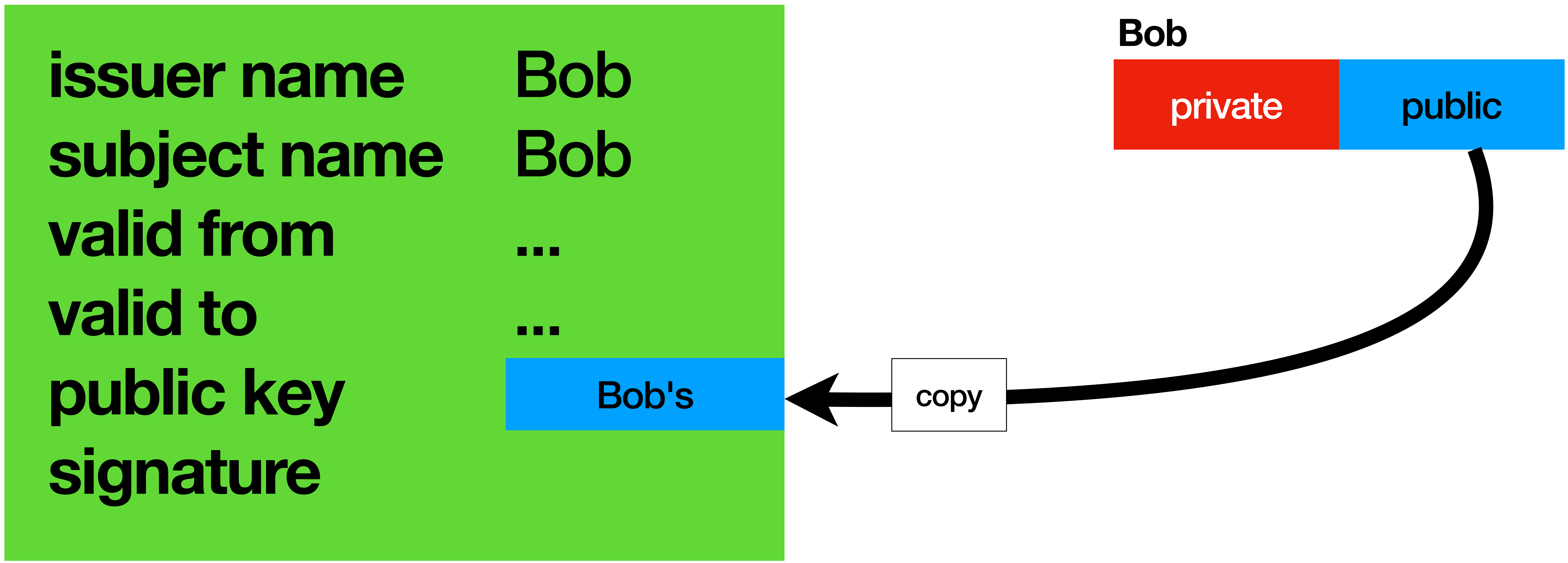
X.509 Self-Signed Certificates

issuer name	Bob
subject name	Bob
valid from	...
valid to	...
public key	
signature	

Bob

private	public

X.509 Self-Signed Certificates



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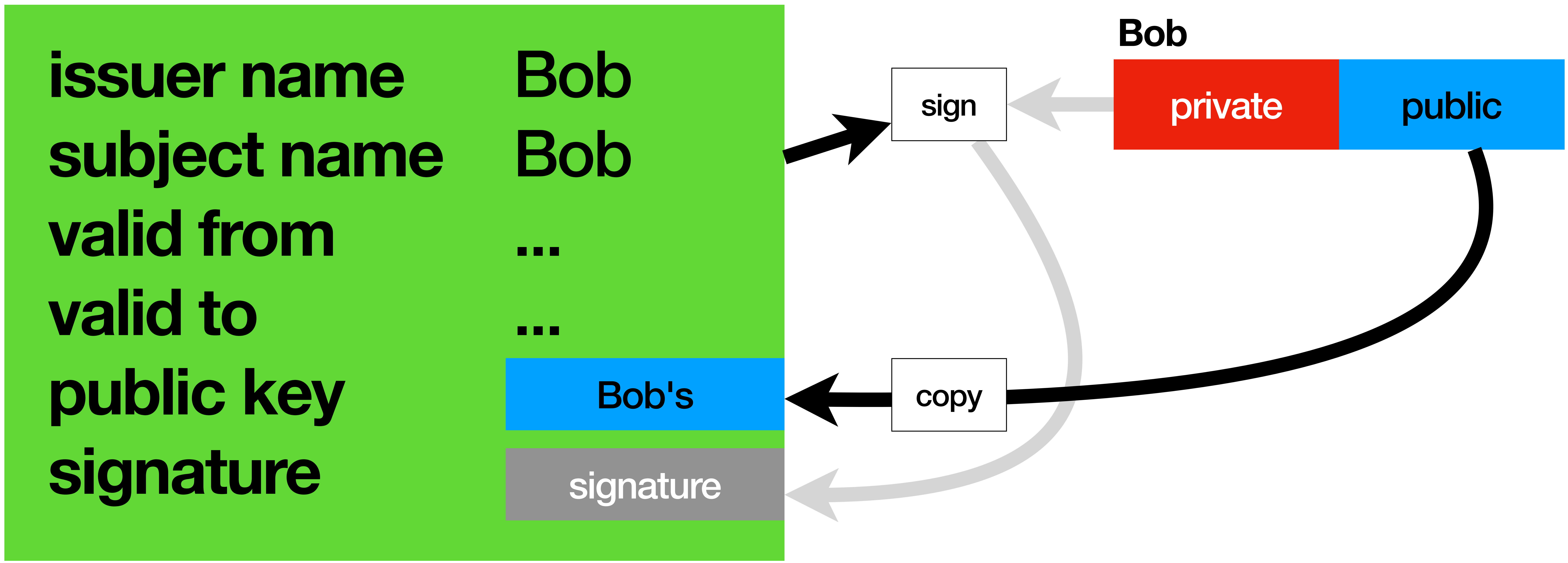

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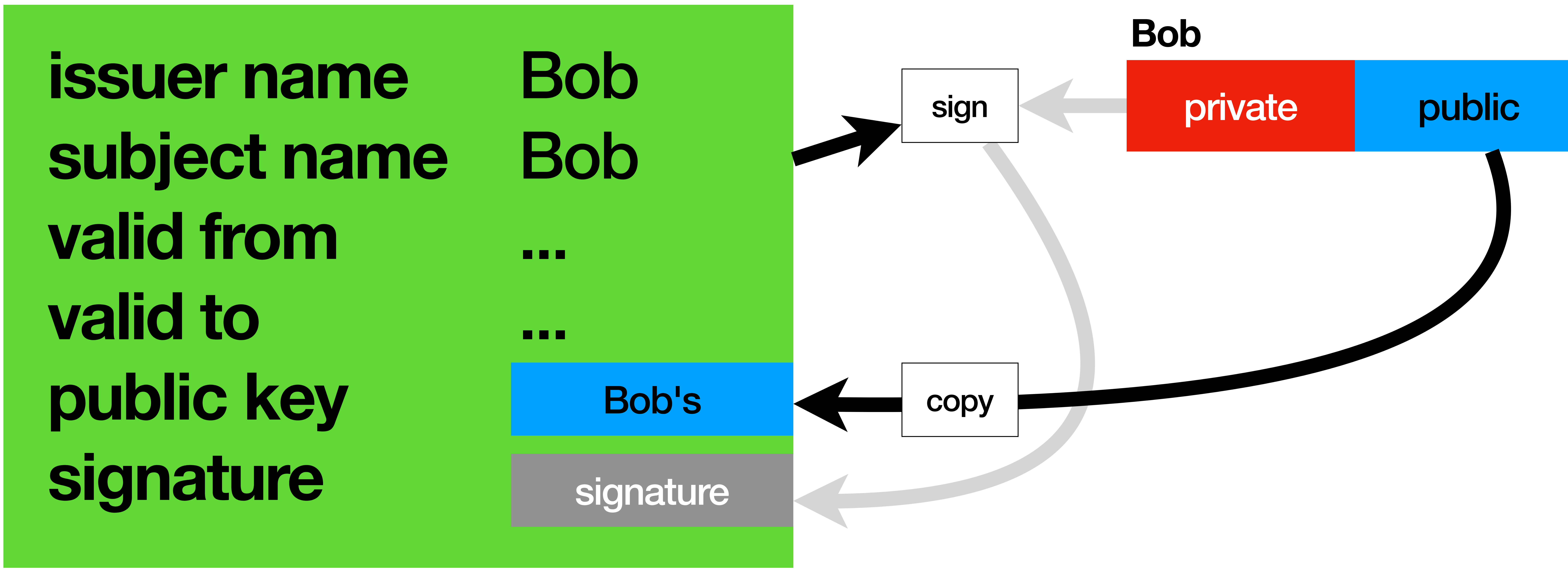

Self-Signed Certificates are **auto proclamations of Identity**

Certificates

Certificates

no way self-signed certificates are enough

Certificate Authorities